Scenario 2: Bad ConfigMap Rollout

Objective

Diagnose a rollout where application configuration and Kubernetes health probes no longer agree.

Trigger the Incident

Run this from the repository root against your own lab environment:

```
./scripts/chaos/break-config.sh student-portal
```

The script saves the current port, changes the ConfigMap, and restarts the student portal backend Deployment. Wait for Datadog and Kubernetes to observe the rollout failure before troubleshooting.

Situation

The student portal has reduced rollout capacity after a configuration release. Existing pods may still serve requests, but new pods do not become Ready.

Symptoms

- New pods fail to become Ready.
- Restart counts may increase while older pods still serve.

What You Know

- Datadog shows unavailable replicas and increasing container restarts for `student-portal-backend`.
- Kubernetes health events began after a rollout.
- The process in each new container appears to start.

Start With Datadog

1. Open **Dashboards > SRE Lab Scenario Signals** with a last-15-minutes window.
2. Filter **Unavailable Replicas** to `kube_namespace:student-portal` and `kube_deployment:student-portal-backend`. Look for a value above zero.
3. In **Container Restarts**, use the same filters and look for a rising `kubernetes.containers.restarts` series.
4. Open **Monitors > Manage Monitors** and inspect **Deployment has unavailable replicas** and **Pod restarts detected**. Expand the `student-portal-backend` alert group.

5. Open **Infrastructure > Kubernetes > Pods** and filter `kube_namespace:student-portal kube_deployment:student-portal-backend`. Compare Ready status and restart counts between older and newer pods.
6. Open **Events > Explorer** with `kube_namespace:student-portal`. Look for readiness/liveness probe failures and BackOff events.
7. Open **Logs > Explorer**, set the last 15 minutes, and search `service:student-portal-backend`. Look for the startup line that reports the listening port.

Record the first unavailable-replica timestamp, newest pod, probe event, restart change, and logged listening port.

Troubleshooting

Check rollout health and identify the newest pods.

```
kubectl get deployments -n student-portal
kubectl get pods -n student-portal
kubectl rollout status deployment/student-portal-backend -n student-portal
kubectl describe pod <pod-name> -n student-portal
kubectl get events -n student-portal --sort-by=.lastTimestamp
kubectl logs <pod-name> -n student-portal
```

After confirming probe failures, compare workload configuration with the probes and rollout history.

```
kubectl get configmaps -n student-portal
kubectl describe configmap student-portal-backend-config -n student-portal
kubectl get deployment student-portal-backend -n student-portal -o yaml
kubectl rollout history deployment/student-portal-backend -n student-portal
```

Important Clues

Connect the probe failures to the application listening port and the port checked by the readiness and liveness probes.

Root Cause

Record the configuration and probe mismatch supported by your evidence. Confirm it with the instructor after the exercise.

Fix

```
./scripts/chaos/break-config.sh student-portal --undo
```

The script restores the saved `PORT` value and restarts the Deployment.

Validation

```
kubectl rollout status deployment/student-portal-backend -n student-portal
kubectl get pods -n student-portal
```

Confirm every pod is Ready, restarts stop increasing, the app works, and both Datadog monitors recover.

DevOps Lesson

Validate configuration and health-probe contracts together before rollout.

STAR Method

Situation

Summarize the partial-capacity rollout.

Task

Explain why only new pods needed investigation.

Action

Describe the Datadog, event, log, ConfigMap, and probe comparison.

Result

State how readiness and monitor recovery were verified.

Natural Spoken Version

Use the matching example in DevOps STAR Scenarios after completing the lab.